THE SQUAMISH CHIEF

3C Hydro is proactive and adaptive, :ompany says of Squamish operations

ınifer Thuncher
ɔv 26, 2019 10:35 AM

:h stranded on Sept. 20 at Road's End on the Cheakamus River. | Chessy Knight

; the chum and coho salmon finish their return to the Cheakamus River, BC Hydro presentativs explain how the company operates in the Sea to Sky and how it tries to reduce the ımber of fish killed because of it.

ter extensive media attention — and public backlash — resulted from a massive fish stranding ıd die-off on the river in September, BC Hydro invited The Chief out to their Squamish ɜadquarters and to the banks of the Cheakamus River to talk to its representatives: Alf Leake, ıvironmental specialist; Darren Sherbot, manager, of operations and planning, and Colin

ɔmbough, natural resource specialist.

ı Sept. 20, local biologists estimate that more than 2,000 pre-spawned pink salmon died from ɛing stranded without enough water in the Cheakamus River due to BC Hydro's operations in e corridor.

ıe BC Hydro reps say a heavy rainstorm meant its operators had to increase flows down the ver to prevent exceeding maximum reservoir levels and ensure public safety.

ɾamatic rainstorms can be a challenge in terms of controlling river flows, the men knowledged.

ımping up usually doesn't cause issues, they say, but ramping down — or reducing the flow to e river — can be tricky.

ɔllowing the September storm, the flows to the river were reduced too drastically, local ɔlogists say.

ıese biologists and corridor fishers have been particularly critical of Hydro's operations when it ımes to killing fish in the Cheakamus and Squamish rivers.

ıe most vocal are members of the Water Use Plan advisory committee. Monitoring Advisory ɔmmittees were established to advise a number of BC Hydro facilities, including in Squamish. ıey include biologists, such as Squamish's Chessy Knight and others with a stake or interest in e local fishery.

C Hydro says when they know better, they do better.

Colin Rombough, natural resource specialist with BC Hydro on the banks of the Cheakamus River. Source: Eli Cornell

)ne thing I feel that hasn't gotten out... is that we have been doing a lot of really good work. We ıve been out monitoring and implementing mitigation every single ramp down for the last year ıd a half now," said Rombough, as the team stood beside the Cheakamus River in Paradise ılley where some of the stranded fish died in September.

/e aren't perfect. Nobody is ever going to be perfect. We are adaptive, we are learning, we are /ing to get there and we are incorporating all of this feedback."

›mbough added he regularly reaches out to stakeholders and the company dialogues with the ıuamish Nation and regulators.

ıe primary fishery regulator is Canadian Fisheries and Oceans (DFO).

: added that since the Sept. 20 fish kill, crews are doing reconnaissance missions before the mpdowns.

he crews are going out and rafting the entire high-risk area of the river," he said, from Road's ıd to the bottom section of the river.

ı that rafting, they are looking and using their knowledge of being out on the river for the last ; years to understand where the risk might lie."

ake added they have also incorporated salvage crews into the mitigation plans.

lvaging involves physically moving the isolated fish from the pools they are stranded in.

ake acknowledged that it isn't always effective and said it is a last resort.

l the reps said there is a lot of co-ordination and planning that goes into every ramp down.

ssons learned from one ramping are incorporated into the next one, they say.

ıe water license for BC Hydro to operate, and control river flows is currently undergoing a ıvernment review.

; Water Use Plan defines the detailed operating parameters to be used by BC Hydro facility anagers in their day-to-day use of the province's water.

he plans are intended to clarify how rights to provincial water resources should be exercised ıd to take account of the multiple uses for those resources," states a BC Hydro online explainer ɔout the plans.

etween 1999 and 2004, BC Hydro prepared 23 Water Use Plans. Squamish's was accepted by e Comptroller of Water Rights and implemented in February of 2006.

ıe Water Use Plan Order Review Program will determine whether the current water-anagement constraints on BC Hydro's operations are achieving its specific environmental and cial objectives and changes can be recommended.

BC Hydro invited The Chief out to their Squamish headquarters and to the banks of the Cheakamus River to talk to their representatives: Alf Leake, environmental specialist (centre); Darren Sherbot, manager, of operations and planning (right), and Colin Rombough, natural resource specialist.Source: Eli Cornell

eview process

ıe Water Use Plan Order Review process has two phases. It starts with a Monitoring Program ·nthesis Report review and that includes a public consultation period. The second phase is the der review phase.

ıe Monitoring Program Synthesis Report summarizes all the monitoring programs over the last) to 12 years, Leake said.

ıat report will be released by the end of this year or early in the new year.

ıen key stakeholders will be consulted, he added.

/here we are taking that information, collecting the monitoring and identifying how we can ake changes to the existing order," Leake said. "We have an existing order, it has ramp rates, it

ıs minimum flows, it has some studies... that is the scope of the things we would look at in the cond phase."

ır some sites, the order review results in the status quo at the BC Hydro facility, such as with ro BC Hydro sites on Vancouver Island, but that isn't likely here, Leake said.

lere, there've been some concerns brought forward, and we recognize that the process is going be more involved. We know that First Nations are going to be interested, we know regulators e interested, we know that there are stakeholders in the community that are involved as well. e're still working out how that engagement is going to happen."

: Hydro moved up the order review period for our area to address issues of concern sooner, ake said.

deo by Chessy Knight

ish kill tipping point?

ıked what number of fish killed would be too many for the company, the Hydro representatives id there isn't a number that is a tipping point.

ut of the Water Use Planning process, we've retained a Fisheries Act authorization for our erations. We did that across the province at all 23 facilities that went through the WUP ocess," Leake. "As part of that fisheries act authorization process, we have conditions. We must onitor and if we monitor and see something we didn't anticipate as a level of impact, then we ve to go through some sort of corrective action process. So, that is how we have been anaging on Cheakamus."

e company works with First Nations, regulators and stakeholders to implement corrective tions.

is not a number. It is just we have this regulatory framework. We have a corrective action ocess we have to go through. And we implement them." he added.

ake said that BC Hydro has "a very close relationship with regulators."

rom some perspectives, that might look like it is not a good thing, but it is a really good thing ecause we are always proactive in coming up with solutions. Even though the Water Use Plan as set 10 or 15 years ago, we have actually made significant changes and gains in several of our cilities as we go through the process so it is really with that framework in place we can actually plement those changes."

ight said that there is no way to know, without better monitoring, what impact the operations e having on fish populations.

think we can all agree here, the eye-opening, jaw-droopingly bad stranding events that we saw summer 2018 and 2019 surely exceed some 'tipping point,'" she said.

eake is correct, BC Hydro does have a very close relationship with regulators... too close."

ıight said given this close relationship, it seems unlikely BC Hydro would ever be charged or ıed for killing fish.

he DFO was unable to confirm or deny that it has never issued fines or charged the utility, prior press deadline.]

Vhy the disconnect with local stakeholders?

ked why there is discord between Squamish fish conservationists and those at BC Hydro, Leake ›inted to the end of some monitoring programs.

Ve were involved in this [Water Use Planning committee] and it was an annual meeting, viewing the results of the studies and checking back with the stakeholders, and regulators, and rst Nations as to whether or not the studies were effective," Leake said.

ake added that when some studies ended, that upset Squamish fish conservationists.

he programs end for two reasons: they run their course, or two, they have answered the ıestions. For the most part, we answered management questions except for two critical studies monitoring program Number 1 that looks at salmon out-migration... and Number 3... — that oks at steelhead productivity."

e said these are two very challenging questions.

o, what is the relationship between flow and productivity? Without major changes to our ›erations, we found over the last 10 years, we can't really tease out that answer."

r BC Hydro, it came down to the choice to keep all monitoring when the company knows it n't answer the management questions under the current Water Use Plan or pausing the onitoring until the Water Use Plan Order Review runs its course.

Ve made sort of a half-and-half decision. On the salmon productivity one, we are going to pause at monitoring and wait for the WUP Order review re-evaluate and hopefully, improve on onitoring terms of reference and study design and implement it there, if that is what the WUP

R decides and the Number 3 [monitoring study,] we said that it is still answering some uestions... so we will keep that going."

r her part, Knight, biologist and WUP committee member, said that as far as the advisory mmittee is concerned, the meetings are to continue, so should not be referred to in the past nse.

he WUP monitoring committee is expecting a meeting this year to review 10-plus year synthesis onitoring reports, the ramping study report, and to discuss next steps for monitoring. No eetings have been scheduled, and no reports have been made available to us. At the very least, C Hydro could tell us when these various reports will be completed, stick to those dates, and let know when we can expect to meet and discuss the reports' findings. The whole process for BC ydro communication with the WUP committee is unorganized," she said.

night doesn't disagree with the assessment that some discord is over Hydro limiting its onitoring.

e says there is no mandate or reason for the monitoring to stop.

or some of the programs, such as the juvenile salmon production and steelhead production dult and juvenile), I firmly believe that these programs should continue throughout the duration the power plant's operation," she said.

Why not follow the DFO guidelines?

night has long argued that BC Hydro should at least follow the recommendations of the heries regulator, Canadian Fisheries and Oceans (DFO), but they don't, for the most part.

sked why Hydro doesn't follow DFO's recommendations for rampdowns for 2.5 centimetres per our, the BC Hydro representatives said they keep asking themselves what is the most effective mprate and what is the best way of mitigating risk.

Ve don't want to — without reason — constrain operations. That does have cost implications nd it does have other flooding implications, but that isn't the primary reason that we are looking . We want to understand what is the relationship between stranding risk and ramping rate in rticular." Leake said.

ydro notes that with the Sept. 20 incident they were actually ramping down at 2.5 cms per hour

ıd that still didn't mitigate for the stranding.

Vhat we see, typically in adults... they are going to spawn in 15 cms of water and then they are ›ing to be attached to those areas that they spawn in, so if we dropped flows, they are going to ·and. It is not the rate in those cases."

ıight said she was pleased to hear Hydro acknowledge that ramping rates can impact revenues.

Ve have asked BC Hydro to share with us the lost revenues resulting from slower ramping rates, ch that we can have the larger social conversation about whether the reduction of fish ·anding is worth the lost revenues," she said.

ıight argued while it is true that sometimes even ramping at a slow rate (e.g., 2.5 cm/hour) nnot prevent fish stranding, she said BC Hydro is not acknowledging the other half of the ·anding problem.

ıat fish kill was caused, Knight says, because BC Hydro seems to have cut river flows in half in ss than one day.

Vhen Cheakamus flows are cut in half, like from 40 cms to 20 cms... a slow ramp rate cannot itigate all fish stranding. Forty cms is already a low flow for the Cheakamus; when it's cut in half, ere is simply not enough wetted habitat for fish."

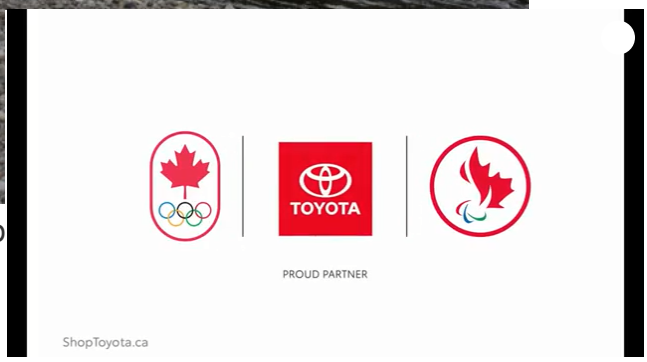
›mbough said it is important to put the DFO recommended rates into context.

hose are the recommended rates in the absence of any other monitoring," he said, adding they e not based on the Cheakamus River.

›ur approach is to implement mitigation and try to understand what is effective on the ıeakamus River. And we think it is a better approach to try to understand what works ·ecifically for this system, rather than just putting blanket instructions down... that may not be fective."

ıight argues the methods used to develop the DFO recommended rates were sound.

Ay own analysis of natural ramping rates on the Cheakamus indicates that a ramp rate of 2.5 ı/hour is a very good starting point," she said.

The Cheakamus River at Road's End, where some p
Cornell

ADVERTISEMENT